

Project Description (What the Application Will Do)

The proposed application is a console-based Car Auction system designed to simulate the process of bidding on vehicles. The application will allow users to view a selection of available cars and participate in an auction by placing bids.

The system will maintain and update the highest bid for each vehicle and determine the winning bidder once the auction concludes. The application will incorporate object-oriented design principles and will utilize arrays to manage multiple vehicles, as well as iteration and decision-making structures to control the bidding process.

2. Target Audience

The intended audience for this application includes individuals interested in vehicle auctions, as well as students and users seeking to understand the basic functionality of an auction system. The application is designed to be simple and accessible for users with minimal technical experience.

3. Application Usage (How It Will Be Used)

The application will operate through a console-based interface where users will be guided through the auction process.

Upon launching the application, users will be presented with a list of available vehicles along with their starting bid prices. The user will select a vehicle and enter a bid amount. The system will evaluate the bid and determine whether it exceeds the current highest bid. If valid, the bid will be accepted and recorded as the new highest bid.

The application will allow repeated bidding through the use of looping structures until the user chooses to exit or the auction ends. At the conclusion of the auction, the system will display the final results, including the highest bid and the winning bidder.

4. Example Scenario (How It Will Work)

A user launches the application and is presented with a list of vehicles available for auction, each with an initial starting bid. The user selects a vehicle and enters their name along with a bid amount.

If the bid is higher than the current highest bid, the system accepts the bid and updates the auction status. Additional users may continue to place bids on the same vehicle. Once bidding is complete, the application announces the highest bidder and the final bid amount for the selected vehicle.